

COIMBRA AP, Portugal

WMO: 085480

Lat: 40.1575N

Lon: 8.4686W

Elev: 172

StdP: 99.28

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.5	(c) 3.7	(d) -4.9	(e) 2.5	(f) 6.3	(g) -2.6	(h) 3.1	(i) 7.1	(j) 8.9	(k) 11.2	(l) 7.9	(m) 11.9	(n) 1.8	(o) 70	(p) 0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 6.1	(c) 5.3	(e) 0.2	(f) 37.6	(g) 1.1	(h) 1.7	(i) -0.6	(j) 38.7	(k) -1.3	(l) 39.7	(m) -1.9	(n) 40.6	(o) -2.7	(p) 41.8		
(a) 7.2	(b) 6.1	(c) 5.3	(e) -1.5	(f) 23.7	(g) 1.4	(h) 0.9	(i) -2.5	(j) 24.3	(k) -3.3	(l) 24.8	(m) -4.1	(n) 25.4	(o) -5.1	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	10.0	10.9	13.2	14.3	16.8	19.5	21.1	21.8	20.5	17.5	12.7	10.6
	DBStd	5.05	2.47	2.59	3.03	3.19	3.28	3.12	2.72	2.71	2.67	2.97	2.72	2.54
	HDD10.0	86	31	19	6	2	0	0	0	0	0	0	6	23
	HDD18.3	1320	259	209	162	129	71	19	3	1	7	52	170	238
	CDD10.0	2191	31	43	104	129	210	285	345	367	317	232	87	43
	CDD18.3	383	0	0	2	7	23	54	90	109	73	25	1	0
	CDH23.3	2959	0	0	17	59	204	464	723	858	496	136	1	0
	CDH26.7	1089	0	0	1	9	67	180	283	341	177	33	0	0
Wind	WSAvg	2.2	2.2	2.2	2.4	2.4	2.3	2.2	2.2	2.2	2.0	2.2	2.2	2.4
Precipitation	PrecAvg	998	136	135	83	87	81	48	13	14	50	103	127	124
	PrecMax	1627	333	337	182	214	176	154	74	60	146	288	467	353
	PrecMin	704	6	28	4	26	0	2	0	0	2	9	2	19
	PrecStd	215	83	90	58	49	49	40	18	16	38	75	86	88
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	21.0	25.6	28.0	32.1	35.2	36.3	37.1	33.9	30.4	22.5	18.3
		MCWB	12.8	13.9	14.5	16.9	19.3	20.6	21.3	21.0	19.7	17.3	14.7	13.9
	2%	DB	15.9	18.3	22.9	25.0	28.3	31.4	32.9	33.6	31.2	26.7	19.9	16.7
		MCWB	12.1	12.2	13.9	16.0	18.2	19.8	20.7	20.3	19.0	17.4	14.6	13.8
	5%	DB	14.7	16.5	20.7	22.4	25.3	28.3	29.7	30.3	28.5	24.3	18.1	15.6
		MCWB	12.1	11.8	13.2	15.3	17.4	19.5	20.2	19.9	18.7	16.9	14.7	13.2
	10%	DB	13.8	15.0	18.4	19.8	22.6	25.6	27.0	27.6	26.0	22.2	16.8	14.5
		MCWB	11.8	11.3	12.5	14.3	16.5	18.7	19.6	19.4	18.3	16.7	14.2	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.2	16.5	18.8	21.3	22.7	23.0	23.3	21.7	20.8	17.7	15.7
		MCDB	15.5	18.1	21.6	24.6	28.1	31.8	31.5	32.4	28.1	23.2	18.7	16.4
	2%	WB	13.7	13.8	15.1	17.1	19.3	21.2	22.0	22.0	20.7	19.7	16.5	14.7
		MCDB	14.5	16.1	19.1	22.8	25.6	28.9	29.7	29.1	27.0	22.6	17.8	15.5
	5%	WB	12.9	13.0	14.2	15.8	18.1	20.1	21.0	21.0	19.9	18.7	15.6	13.7
		MCDB	13.9	15.0	17.9	21.1	24.0	26.7	28.1	27.5	25.6	21.8	17.0	14.8
	10%	WB	12.1	12.1	13.4	14.7	17.0	19.0	20.1	20.1	19.1	17.8	14.6	12.9
		MCDB	13.3	14.3	16.9	18.7	21.6	24.4	26.4	26.1	24.1	20.7	16.1	14.1
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.7	8.6	8.8	9.7	10.6	11.6	12.1	11.1	8.6	6.9	6.4
		MCDBR	7.3	9.7	12.2	12.8	14.2	15.5	16.5	16.3	15.0	12.0	8.0	6.7
		MCWBR	5.0	5.7	6.4	6.4	7.0	6.8	6.6	6.3	5.9	5.5	4.8	4.6
	5% WB	MCDBR	5.3	7.6	9.6	11.5	12.2	13.6	14.1	13.3	11.9	8.8	5.8	5.3
		MCWBR	4.2	5.1	5.6	6.3	6.6	6.5	6.3	6.0	5.5	4.8	4.3	4.4
Clear-Sky Solar Irradiance	taub	0.323	0.336	0.368	0.368	0.375	0.379	0.368	0.384	0.373	0.360	0.333	0.320	
	taud	2.516	2.473	2.389	2.398	2.403	2.408	2.423	2.379	2.434	2.488	2.521	2.523	
	Ebn at Noon	843	882	884	906	903	896	903	879	868	841	823	816	
	Edh at Noon	75	90	109	115	117	116	114	116	103	88	74	70	
All-Sky Solar Radiation	RadAvg	2.02	3.03	4.08	5.28	6.45	7.10	7.32	6.61	5.13	3.37	2.26	1.83	
	RadStd	0.32	0.42	0.55	0.61	0.60	0.46	0.31	0.27	0.40	0.37	0.37	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page